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Annexe 2
Annex 2

Valeurs recommandées des fréquences étalons destinées à la mise en pratique de la définition du mètre et aux représentations secondaires de la seconde

Recommended values of standard frequencies for applications including the practical realization of the metre and secondary representations of the second

Partie 2
Part 2

Rayonnements spectraux de la lampe

Spectral lamp radiations

CCL-CCTF Frequency Standards Working Group

Bureau International
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Part 2: Spectral lamp radiations

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Appendix 2

1. CIPM recommended values

Vacuum wavelengths, λ, for ^{86}Kr, ^{198}Hg and ^{114}Cd transitions

Atom	Transition	λ/ pm
^{86}Kr	$2p_9 - 5d'_4$	645 807.20
^{86}Kr	$2p_8 - 5d_4$	642 280.06
^{86}Kr	$1s_3 - 3p_{10}$	565 112.86
^{86}Kr	$1s_4 - 3p_8$	450 361.62
^{198}Hg	$6^1P_1 - 6^1D_2$	579 226.83
^{198}Hg	$6^1P_1 - 6^3D_2$	577 119.83
^{198}Hg	$6^3P_2 - 7^3S_1$	546 227.05
^{198}Hg	$6^3P_1 - 7^3S_1$	435 956.24
^{114}Cd	$5^1P_1 - 5^1D_2$	644 024.80
^{114}Cd	$5^3P_2 - 6^3S_1$	508 723.79
^{114}Cd	$5^3P_1 - 6^3S_1$	480 125.21
^{114}Cd	$5^3P_0 - 6^3S_1$	467 945.81

For ^{86}Kr, the above values with a relative expanded uncertainty $U = 2 \times 10^{-8}$, where $U = ku_c$ $(k = 3)$, apply to radiations emitted by a hot-cathode discharge lamp containing ^{86}Kr, of a purity not less than 99 %, in sufficient quantity to assure the presence of solid krypton at a temperature of 64 K, this lamp having a capillary with an inner diameter from 2 mm to 4 mm and a wall thickness of about 1 mm.

It is estimated that the wavelength of the radiation emitted by the positive column is equal, to within 1 part in 10^8, to the wavelength corresponding to the transition between the unperturbed levels, when the following conditions are satisfied:

- the capillary is observed end-on from the side closest to the anode;
- the lower part of the lamp, including the capillary, is immersed in a cold bath maintained at a temperature within one degree of the triple point of nitrogen;
- the current density in the capillary is (0.3 ± 0.1) A$\,$cm^{-2}.

For ^{198}Hg, the above values with a relative expanded uncertainty $U = 5 \times 10^{-8}$, where $U = ku_c$ $(k = 3)$, apply to radiations emitted by a discharge lamp when the following conditions are met:

- the radiations are produced using a discharge lamp without electrodes containing ^{198}Hg, of a purity not less than 98 %, and argon at a pressure from 0.5 mm Hg to 1.0 mm Hg (66 Pa to 133 Pa);
- the internal diameter of the capillary of the lamp is about 5 mm, and the radiation is observed transversely;
- the lamp is excited by a high-frequency field at a moderate power and is maintained at a temperature less than 10 °C;
- it is preferred that the volume of the lamp be greater than 20 cm^3.

For ^{114}Cd, the above values with a relative expanded uncertainty $U = 7 \times 10^{-8}$, where $U = ku_c$ $(k = 3)$, apply to radiations emitted by a discharge lamp under the following conditions:

- the radiations are generated using a discharge lamp without electrodes, containing ^{114}Cd of a purity not less than 95 %, and argon at a pressure of about 1 mm Hg (133 Pa) at ambient temperature;
- the internal diameter of the capillary of the lamp is about 5 mm, and the radiation is observed transversely;
- the lamp is excited by a high-frequency field at a moderate power and is maintained at a temperature such that the green line is not reversed.

2. Source data

The recommended wavelengths are those recommended by the CIPM in 1963 [1], [2].

^{86}Kr, ^{198}Hg, ^{114}Cd, spectral lamp radiations

References

[1] *BIPM, Com. Cons. Déf. Mètre*, 1962, **3**, 18-19

[2] *BIPM, Proc. Verb. Com. Int. Poids et Mesures*, 1963, **31**, 26-27.